

6. Connect this with the Box Plot, Linear Correlation and X-Partitioner nodes.
7. In the Box Plot, exclude all the columns except Calories and visualise the plot. We can visualise the robust statistical parameters of minimum, lower quartile, median, upper quartile, and maximum value.
8. The Linear Correlation node calculates a correlation coefficient for each pair of selected columns, i.e., a measure of the correlation of the two variables. Here, all columns are included to calculate the correlation coefficient.
9. The X-Partitioner node is used for performing the cross-validation in the data set. This node is the first in a cross-validation loop. At the end of the loop, there must be an X-Aggregator to collect the results from each iteration. All nodes in between these two nodes are executed as many times as iterations should be performed. We perform 10 validations using random sampling, and the best out of these 10 is given as the output.
10. Connect the first partition with the Random Forest Learner node. Select the Target column as 'Calories burn' and let the rest of the options be the same as the default. Select the split criteria as Information Gain Ratio.
11. Now, connect the Random Forest Predictor node with the second partitioning and the Random Forest Learner node.
12. Connect this node with the X-Aggregator node. This node collects the result from a Predictor node, compares the predicted class and real class, and outputs the predictions for all rows and the iteration statistics. Select the Target column as 'Calories burn' and the Prediction column as 'Prediction (Calories burn)'.
13. Connect this node with the Scorer and Bar Chart nodes.

14. In the Scorer node, select the first column as 'Calories burn' and the second column as 'Prediction (Calories burn)', and save it.
15. In the bar chart, enter the column as 'Prediction (Calories burn)',

name the chart, and then save it. Finally, execute all the nodes and view the output.

The complete flow for calorie burn prediction using a random forest predictor is given in Figure 1.

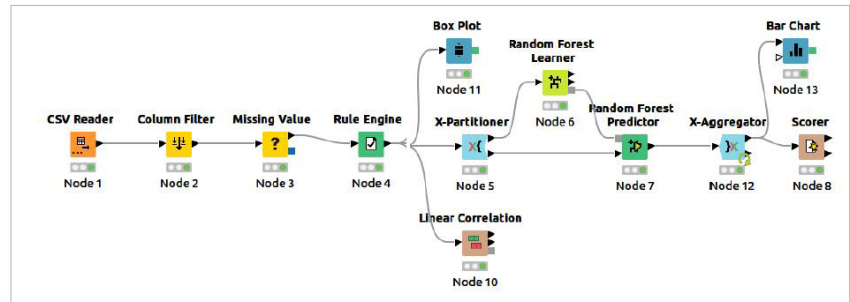


Figure 1: KNIME flow for calorie burn prediction using a random forest predictor

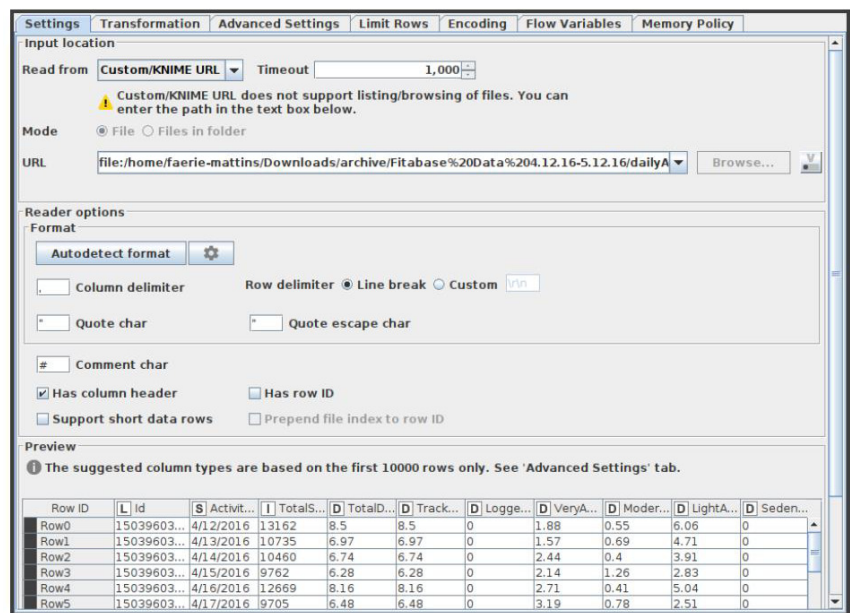


Figure 2: Configuration of CSV Reader node

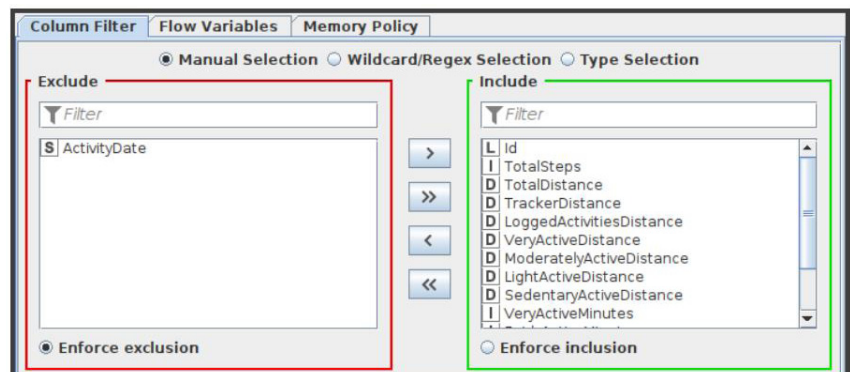


Figure 3: Configuration of Column Filter node